

Up: Delivering digital banking to 100,000+ Australians with Google Cloud

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ABOUT UP

With Google Cloud, Up has grown to 100,000 customers in eight months by providing a secure, responsive, rapid digital banking app.



About Up

Up—founded in July 2017 as a collaboration between Bendigo and Adelaide Bank and software development firm Ferocia—is one of Australia's leading digital banks. The bank has been in full production since October 2018 and in the eight months following launch grown to more than 100,000 customers. Up gives customers visibility into when, where, and how they spend their money; predicts upcoming bills, helping customers pay them on-time; automatically generates reports for financial insights; provides competitive interest across

Google Cloud results

- Delivers availability of 99.99%+

Support
to 100,000+

- Enables five deployments per day to improve products and customer experiences
- Completes full overseas replication for disaster recovery in 90 minutes

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eight |

multiple saver accounts as well as offering no monthly fees or minimum deposits; does not charge international purchase fees online or in-store; and provides smart push notifications, and integration with a virtual assistant.

Industries: Financial Services & Insurance

Location: Australia

(<https://cloud.google.com/kubernetes-engine/>),
Google Cloud (<https://cloud.google.com/>),

Cloud Storage

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

Digital banks—also known as neobanks—are transforming Australia’s financial landscape. These banks are made available via mobile apps and are not bound to use traditional processes employed by traditional banks.

Cloud SQL (<https://cloud.google.com/sql/>)

, Cloud Load Balancing

Cloud Storage (<https://cloud.google.com/storage/>)

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Operations

(<https://cloud.google.com/products/operations/>)

Up (<https://up.com.au/>), founded in 2017, is a collaboration between Bendigo Bank and software development firm

Cloud Load Balancing

(<https://cloud.google.com/load-balancing/>)

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Australia’s leading digital banks. “We have been in full production since October 2018 and in the eight months following launch, grown to more than 100,000 customers,” says Michael Morris, Head of Technology at Up. “People are responding enthusiastically to our transparency, value, agility, technology and our brand ethos.”

Up gives customers visibility into when, where, and how they spend their money; predicts upcoming

bills, helping customers pay them on-time; automatically generates reports for financial insights; provides competitive interest across multiple saver accounts as well as offering no monthly fees or minimum deposits; does not charge international purchase fees online or in-store; and provides smart push notifications and integration with a virtual assistant.

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—Michael Morris, Head of
Technology, Up

A technology company making a bank

Morris—who also heads technology at Ferocia and is responsible for developing Bendigo Bank’s award-winning mobile and internet banking platform, the fifth largest retail banking platform in Australia—describes Up as “a technology company making a bank, rather than a bank making technology.”

“This means we approach most problems with a technology mindset and technology-based solutions,” he says.

Up aimed to become Australia’s first cloud-based retail bank and carefully evaluated potential cloud service providers. “We needed a cloud provider to help us clear potential obstacles to moving at speed and address the security and regulatory requirements of launching a new financial services business,” says Morris.

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Technology, Up*

Meeting regulatory requirements

Ferocia and Bendigo and Adelaide Bank created the first, non-public iteration of Up on a multinational cloud service. However, launch of the Google Cloud Sydney region in 2017—enabling Google Cloud customers to run workloads and store data in Australia—prompted the businesses to move Up to the platform. "Moving to Google Cloud in Sydney would help us meet the strict regulatory requirements imposed by the Australian Prudential Regulation Authority, which supervises the country's financial services institutions," says Morris. "We also wanted to take advantage of the native Kubernetes support available through Google Cloud."

Securing customer and corporate data and systems is also a business-critical issue for Up. The digital bank was satisfied by Google Cloud's end-to-end security model, backed by an information security team comprising experts in information, application, and network security.

Up elected to move to a Google Cloud architecture centered on the [Google Kubernetes Engine](https://cloud.google.com/kubernetes-engine/) (<https://cloud.google.com/kubernetes-engine/>) managed, production-ready environment for deploying containerized applications. Up is also using [Cloud Storage](https://cloud.google.com/storage/) (<https://cloud.google.com/storage/>) to store and archive data; [Cloud SQL](https://cloud.google.com/sql/) (<https://cloud.google.com/sql/>) to run relational databases; and custom instrumentation with [Stackdriver](https://cloud.google.com/stackdriver/) (<https://cloud.google.com/stackdriver/>) for monitoring and logging. The business also uses [Cloud Load Balancing](https://cloud.google.com/load-balancing/) (<https://cloud.google.com/load-balancing/>) to load-balance compute resources to meet strict availability requirements.

The Google Cloud infrastructure and Up systems connect to Bendigo and Adelaide Bank's on-premises and cloud services systems.

Winning support from customers

Running on Google Cloud has enabled the fledgling digital bank to deliver the experience required to win support and business from customers of all ages and meet demanding internal requirements. The Up team is undertaking five or more deployments per day to improve the products and services customers are using—well ahead of the one deployment per quarter Morris says occurs at many traditional banks.

"The 'cost' of a deployment at a traditional bank can be up to seven hours' downtime, whereas with Google Cloud, we don't experience any downtime at all during a deployment," says Morris. "So we can

enhance features, fix bugs, and ship new products as fast as we need to.”

"We see our journey as growing the breadth and depth of our customers in Australia.

Google Cloud is supporting this aim by giving us the ability to deploy change safely, securely, and at speed, so we can focus on delighting customers.”

—Michael Morris, Head of Technology, Up

Security a key focus

Up is also reaping the rewards of Google Cloud's focus on security, and Morris notes the speed with which the team applied patches to protect the digital bank against malware known as Spectre and Meltdown targeting processor vulnerabilities. The Up team itself is using a custom app tester and workflow based on [Binary Authorization](https://cloud.google.com/binary-authorization/) (<https://cloud.google.com/binary-authorization/>) on Google Kubernetes Engine to improve the security and trust of container images. "This enables us to work with containerized apps safely as well as quickly," says Morris.

Up is using scripts and tools to automate the creation of environments—providing additional security and improving test and development. "We run multiple test and development environments, as well as production—full automation enabled through Google Cloud ensures we don't neglect security when creating them," explains Morris. "We back the automation with processes that ensure our configurations are audited, peer-reviewed, and undertaken carefully."

20x replicated copies several times a day

The digital bank complements these measures with a testing regime that spins up 20 replicated copies of its infrastructure several times a day to ensure that safety and security is being maintained. Google Cloud also enables Up to complete full system regression tests in 26 minutes, spin its full system up in 45 seconds, and complete full overseas replication for disaster recovery in just 90 minutes.

Uptime is also key to a new venture that needs to maintain customer confidence to succeed.

According to Morris, Up has recorded only 35 minutes downtime—planned or unplanned—since launch in October 2018, delivering availability well over 99.99%.

Running on Google Cloud enables Up to manage its Android and iOS apps plus server-side components with a team of about 40 people, 25 of whom are engineers. “With Google Cloud, we are leaner and more agile than some major traditional banks that have 150-person-strong teams managing the client-side Android and iOS apps alone,” says Morris. “We have a technology company’s continual focus on getting better.”

This focus has enabled Up to deliver a number of innovative features, including an in-app signup process that takes less than three minutes and a “conversational payments” feature that enables customers to make payments in real time to other people and businesses. According to Morris, innovation and strong advocacy from customers are key contributors to Up’s growth.

ML being considered

With Up running smoothly, the business is now considering a range of Google Cloud products and features. “Up is on some of the beta programs for security health analytics, threat detection, and binary authorization,” says Morris. “We’re also interested in the potential of machine learning to assist customer support and fraud detection.”

Up is now poised for a bright future, and Morris points to the experience of digital banks in the United Kingdom that, according to Accenture, are on track to triple customer numbers to 35 million in the next 12 months. "We see our journey as growing the breadth and depth of our customers in Australia," says Morris. "Google Cloud is supporting this aim by giving us the ability to deploy change safely, securely, and at speed, so we can focus on delighting customers."

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business.

